

LEAD DEVELOPER & IDEA: Rashid Ali	ACADEMIC PROGRAM: AI Web Development (DTAN)	INITIATIVE: Balochistan Youth Empowerment	LIVE DEPLOYMENT: homefixquetta.infinityfree.io
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1. The Business Problem & Market Context

Quetta, the provincial capital of Balochistan, has experienced rapid urban and residential expansion across sectors such as Satellite Town, Jinnah Town, Cantt, Zarghoon Road, Samungli, and Airport Road. Despite this growth, the local residential repair and maintenance sector has remained entirely unorganized, informal, and fragmented.

Key Urban Challenges in Quetta

- Arbitrary & Opaque Pricing:** Freelance technicians quote inconsistent rates on-site with zero standardization, written estimates, or service guarantees.
- Unverified Freelance Workforce:** Lack of background checks, identity screening (CNIC), or safety verification for in-home service appointments.
- Severe Emergency Friction:** High difficulty in finding qualified plumbers during winter freeze/tank bursts or electricians during power load shedding.

Economic & Social Opportunity

- Skilled Workforce Empowerment:** Formalizing freelance income for local Balochistan technicians with a steady, dignity-centered job order pipeline.
- Digital Neighborhood Dispatch:** Overcoming geographic dispatch latency across dispersed Quetta sectors via digitized service hubs.
- Transparent Standard Rates:** Offering upfront fixed pricing in Pakistani Rupees (PKR) backed by a 30-day workmanship warranty.

2. The Solution Vision & Ecosystem

HomeFix Quetta was conceived, designed, and engineered by **Rashid Ali** as a high-performance, two-sided on-demand digital platform connecting homeowners with certified local technicians across 4 mission-critical trades:

 **Plumbing**

Pipe leakage repairs, sanitary fittings, overhead tank deep cleaning, water pump installations.

 **Electrical & Solar**

UPS inverter wiring, DB breaker load balancing, solar panel setup, short-circuit diagnostics.

 **Painting**

WeatherShield exterior coatings, interior emulsion, roof & ceiling seepage proofing.

 **Handyman**

LED TV wall mounting, door lock replacement, curtain rod fitting, appliance mounting.

Core Platform Value Proposition

A modern web application offering instant AJAX live catalog search, multi-step booking with photo upload preview, public live tracking (Ref #), verified customer testimonials, and an administrative operations control center.

3. Technology Stack & Multi-Tier Architecture

Frontend & Presentation Layer

- Tailwind CSS 3.4:** Utility-first responsive design framework customized with brand tokens.
- GSAP 3.12 & ScrollTrigger:** Hardware-accelerated animations and scroll-based micro-interactions.
- Leaflet.js 1.9 & OpenStreetMap:** Interactive Quetta dispatch hub geolocation map.
- HF JavaScript Engine:** Global non-blocking AJAX module, toast dispatcher, and form validators.

Backend & Data Persistence Layer

- PHP 8.0+ Runtime:** OOP business logic, session guards, and REST-style AJAX controllers.
- MySQL 8.0+ / MariaDB:** Relational storage with InnoDB engine, foreign keys, and indexes.
- Singleton Database PDO Layer:** Centralized connection pooling and prepared SQL execution.
- Apache .htaccess Hardening:** Extensionless clean routing and directory access restrictions.

4. Relational Database Schema & Data Models

Table	Keys & References	Key Attributes & Purpose
users	id (PK)	name, email, phone, password_hash, role (admin/customer), avatar, area, status
categories	id (PK)	name, slug, description, icon, image, status (4 Core Service Sectors)
services	id (PK), category_id (FK)	name, slug, price, duration, image, includes_list, is_featured, status
technicians	id (PK)	name, email, phone, specialty, experience_years, rating, availability, status
bookings	id (PK), service_id, tech_id (FK)	booking_ref, customer_name, area, address, date, time, status, is_viewed, total_amount
reviews	id (PK), booking_id (FK)	customer_name, rating (1-5), review_text, status (approved/hidden)
contact_messages	id (PK)	name, email, phone, subject, message, is_read (0/1), created_at

5. Defensive Engineering & Security Controls

SQL Injection Elimination

100% prepared PDO statements with EMULATE_PREPARES=false. Zero raw string queries.

BCRYPT Cryptography

Passwords hashed via PASSWORD_BCRYPT and verified with constant-time password_verify().

Multi-MIME Upload Guard

Binary file header validation via PHP finfo, 5MB limit, randomized alphanumeric filenames.

AI Tools, Project Impact & Future Roadmap

AI Engineering Integration, Community Impact & Scalability

AI TOOLS & IMPACT

6. AI Tools & Modern Engineering Methodologies

As part of the **AI Web Development** specialization, modern generative AI tools and agentic coding workflows were strategically integrated throughout the project lifecycle:

1. AI Full-Stack Scaffolding

Leveraged Google Gemini AI models for architectural design, normalized database schema generation, secure PDO queries, and modular PHP controllers.

2. UI / Motion Engineering

AI-assisted tokenization of Tailwind CSS color palettes, GSAP cubic-bezier animation curves, and responsive glassmorphism navbar styling.

3. Automated QA & Auditing

Used AI-driven static analysis to audit SQL foreign key integrity, resolve session race conditions, and implement real-time notification states.

7. Deployment & Socio-Economic Impact

Production Deployment Matrix

- **Production URL:** homefixquetta.infinityfree.io
- **GitHub Repository:** github.com/Dopevibe/homefix-quetta
- **Zero-Cost Architecture:** Full cloud PHP/MySQL deployment with custom SSL certificates.
- **Performance:** Sub-second page loads optimized via CSS/JS minification and asset compression.

Community Impact for Balochistan

- **Economic Inclusion:** Formalizes freelance income for skilled workers across Quetta.
- **Consumer Trust:** Introduces verified reviews and fixed pricing to a previously unregulated market.
- **Technological Advancement:** Demonstrates practical, localized digital solutions created by Balochistan youth.

8. Future Scalability & Enhancements

1. AI Automated Dispatch

Machine learning model matching technicians based on GPS proximity and trade rating.

2. Multi-Lingual Support

Localization into Urdu, Pashto, and Brahui for wider accessibility.

3. WhatsApp Bot API

Automated real-time booking confirmation and status alerts via WhatsApp Business API.

PROJECT AUTHOR & SUBMITTER

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